Unit 1 - Professional Communication

1.1 Fundamentals of Communication:

Definition of Communication:

Communication is the process of exchanging information, ideas, thoughts, or feelings through verbal, nonverbal, or written means. It is fundamental in all human interactions, including in professional settings such as engineering.

Popular Definitions:

M.L.K. Fischer:"Communication is the process of sending and receiving messages, which may or may not have meaning."

Stephen P. Robbins:"Communication is the transmission of meaning from one person to another."

Shannon and Weaver (1949):"Communication is a process of transmitting information from a source to a destination through a medium."

Why is Communication Important?

Building Relationships: Good communication is essential for establishing and maintaining professional relationships.

Effective Collaboration: In engineering, communication is key to teamwork, sharing ideas, and achieving common goals.

Problem Solving: Communication helps in understanding issues clearly and coming up with solutions.

Professionalism: Clear communication demonstrates competence and professionalism, essential in all fields, especially engineering.

Linear Model of Communication

Definition: The **Linear Model of Communication** is a simple one-way communication model where a message travels from the sender to the receiver with no feedback loop. This model focuses on the message itself, the sender, and the receiver.

Key Elements:

Sender: The individual who initiates the message.

Message: The information being communicated.

Channel: The medium through which the message is sent (e.g., email, speech, written document).

Receiver: The person who receives the message.

Noise: Any interference or barrier that distorts the message (e.g., background noise, misunderstandings).

Shannon and Weaver Model of Communication

Definition: The **Shannon and Weaver Model** expands on the Linear Model by adding an element of noise and introducing the concept of encoding and decoding, making it a more realisticrepresentationofcommunication.Thismodelisoftenreferredtoasthe**"Motherof All Models"** because it is foundational in the study of communication.

Key Elements:

Sender (Information Source): The originator of the message or information.

Encoder (Transmitter): Converts the message into a form that can be transmitted (e.g., speaking or writing).

Channel: The medium through which the encoded message is transmitted (e.g., voice, email).

Noise: Any disturbance that interferes with the transmission or reception of the message.

Decoder (Receiver): The person or system that interprets the message.

Receiver (Destination): The person who receives and decodes the message.

1.2 Characteristics of Language

- **Definition of Language:**
  - **Language** is a system of symbols, sounds, and rules used for communication. In professional communication, language allows engineers to express complex ideas, share technical information, and collaborate effectively with colleagues, clients, and stakeholders.

Key Characteristics of Language

- **Clarity:**
  - **Definition:** Language must be clear and precise to ensure the message is understood exactly as intended.
  - **Example:** When writing a project report, engineers need to use clear language to explain technical processes so that stakeholders can follow the steps without confusion.
- **Conciseness:**
  - **Definition:** Language should be brief and to the point, avoiding unnecessary words or redundancies.
  - **Example:** Engineers often have to communicate technical data and updates. Using concise language helps the audience quickly grasp the key points.

According to the ideas put forward by eminent linguists:

Language is Artificial

Language is created by people. It does not exist in isolation or outside the minds of people. It is created by human as they need it. Every symbol is attached to a particular thought or thing, called a referent which is created by humans. That's why language is Artificial.

Language is Abstract

Language is abstract as it represents generalized ideas of things or thoughts. The idea which the word represents is different every time. A **'table'** can be of different shapes and sizes, and still be called a table. This happens because meanings get associated with symbols and users keep expanding the range of meanings.

Language is Arbitrary

There is no direct relationship between a word and the idea or object it represents because language keeps on changing to include new concepts and words can attach a number of specific and arbitrary meanings.

Language is Restricted

When we translate our thoughts into language, some meaning is lost in the process. No symbol or word can transmit the exact reality. That is one reason you sometimes find yourself saying that you cannot find words to express your feelings. This is because language is restricted. In other words, it has limitations.

Language is Recursive

Recursiveness is the characteristic of language which enables you to generate any number of sentences using the same basic grammatical templates. It also allows you to express any idea, thought or feeling using the same finite vocabulary. It is a characteristic of language which enables one to generate any number of sentences using the basic grammatical templates. It also allows one to express any idea, thoughts, or feeling using the same vocabulary. These basic characteristics of language make it an effective means of communication.

Language is Creative

Language is very creative and it can create wonder. Every year number of words can be added; taken from different languages through the following processes:

1. Borrowing - taking over words from other languages like **'alcohol'** form **'Arabic'**
2. Constructing Portmanteau Words – words made by combining the sound and meaning of two different words (**netiquette = net + etiquette**)
3. Back Formation – where a word of one type, usually a noun is reduced to a word of another type, usually a verb like **'opt'** from **'option'**

1 .3 Levels of Communication:

1. Intrapersonal Communication

This takes place within the individual. For example when you **"feel hot"**, the information is sent to brain and you may decide to **"turn on the cooler"**, responding the instructions sent from brain to hand. Here relevant organ is sender, electrochemical impulse is message and brain is receiver. Next the brain assumes the role of sender and sends the feedback that you should switch on the cooler. So this process can be termed as intrapersonal communication.

2. Extrapersonal Communication

It is a communication between human beings and non human entities. This requires perfect coordination between sender and receiver. When your pet dog comes to you wagging its tail; as soon as you return home is an example of extrapersonal communication.

3. Interpersonal Communication

It is a sharing of information among people. It includes a few participants who are close to one another. Here many sensory channels are used and immediate feedback can be obtained. Also, the roles of sender and receiver keep alternating. Interpersonal communication can be formal and informal.

4. Organizational Communication

Communication in an organization takes place at different levels. Since a large number of employees are involved in several activities, the need to communicate becomes greater in an organization. This communication can further be divided into:

Internal – operational

External – operational

Personal

5. Mass Communication

There are several mass media such as journals, television, newspapers, internet which mediate such communication to the large audience. Here audience is heterogeneous and anonymous with impersonal approach. This type of communication is more persuasive in nature than any other form of communication and requires utmost care on the part of sender.

1.4 Flow of Communication

1. Downward Communication

Downward communication flows from a manager down the chain of command. When he informs, instructs, or advises his subordinates, the communication flows in a downward pattern. It is used to convey routine information, new, policies or procedures, etc through memos, notices, and interactions and so on.

2. Upward Communication

When subordinates send reports to their superiors, the communication flows upward. This type of communication keeps managers aware of how employees feel about their jobs, colleagues and organization. Managers rely on upward communication form making certain decisions or solving problems.

3. Lateral or Horizontal Communication

This form of communication takes place among peer groups or hierarchically equivalent employees. Such communication is often necessary to facilitate coordination, save time, and bridge the communication gap among various departments.

4. Diagonal or Cross-wise Communication

This flows in all directions and cuts across functions and levels in an organization. This process is quick and efficient. When a sales manager communicates directly with director, they set an example of diagonal communication. The increased use of emails encourages cross-wise communication. There is no doubt that it is quick and efficient.

1.5 Barriers of Communication

Definition of Communication Barriers:

Communication Barriers are obstacles that interfere with or prevent the effective exchange of information. These barriers can occur at any stage of communication and can distort or obstruct the intended message.

Why Are Communication Barriers Important to Understand?

Engineers need to be aware of these barriers to ensure that messages are

Delivered accurately, especially in complex and technical fields where clarity is essential.

Identifying and overcoming communication barriers helps improve teamwork, prevent errors, and enhance productivity.

Types of Communication Barriers

- **Physical Barriers:** Definition: These are external factors that physically prevent effective communication, such as noise, distance, or environmental distractions.

Importance in Engineering: Physical barriers are often encountered in noisy work environments or large construction sites, where clear verbal communication can be hindered by background noise.

- **Semantic Barriers:**

  Definition: These barriers occur when words or phrases are misinterpreted due to differences in meaning or understanding. This is often due to technical jargon, language differences, or ambiguous terms.

  Importance in Engineering: In engineering, where precise language is crucial, using terms that are not universally understood can lead to confusion or errors. An engineer trying to give instructions at a noisy construction site where machinery sounds drown out the voice.

- **Emotional Barriers:**

  Definition: Emotional barriers arise when emotions, personal biases, or attitudes interfere with how information is delivered or received. These barriers can prevent people from listening effectively or responding rationally.

  Importance in Engineering: Emotional barriers can arise when engineers or team members are frustrated, stressed, or upset, affecting how they communicate with others. An engineer who is upset about a project delay may communicate in a harsh tone, leading to misunderstanding or conflict with team members.

- **Psychological Barriers:**

  Definition: Psychological barriers occur due to mental factors such as stress, anxiety, or cognitive biases that affect how information is processed.

  Importance in Engineering: Psychological barriers can prevent effective listening, problem-solving, or decision-making, especially under high pressure or tight deadlines. An engineer who is stressed about meeting deadlines may misinterpret feedback during a meeting, leading to miscommunication.

1.6 Nonverbal Communication:

Nonverbal Communication refers to the transmission of messages or information without the use of words. This includes facial expressions, gestures, body language, tone of voice, posture, eye contact, and even the use of space and time. Effective nonverbal communication ensures that engineers and teams can collaborate smoothly, manage stress, and maintain positive client relationships. Nonverbal communication is essential when presenting ideas, giving feedback, or collaborating with colleagues and clients, especially when working in cross-functional or multicultural teams.

Paralinguistic Features

Paralinguistic features are non-verbal cues that help you to give urgency to your voice. Your voice is your trademark. Therefore it is useful to understand the characteristic nuances of voice.

1. Quality

Each one of us has a unique voice and its quality depends upon its resonating mechanism. It may be rich and resonant, soft and alluring, thin and nasal, hoarse and husky, or harsh and irritating. One can make conscious efforts to improve one's quality of voice.

2. Volume

Volume is the loudness or the softness of the voice. Your voice should always project but need not always be loud. You should vary your volume so as to make your voice audible and clear.

3. Pace/Rate

Rate is the number of words which you speak per minute. It varies from person to person and from 80 to 250 words per minute. The normal rate is from 120 to 150 words. Cultivate your pace so as to fit in this reasonable limit. Use pauses to create emphasis. A well paced, varied message suggests enthusiasm, self-assurance and awareness of audience.

4. Pitch

Pitch refers to the number of vibrations per second of your voice. The rise and fall of the voice conveys various emotions. A well balanced pitch results in a clear and effective tone.

5. Articulation

Speakers should be careful not to slop, slur, chop, truncate, or omit sounds between words or sentences. If all the sounds are not uttered properly, the flow of understanding gets interrupted and deters the listener from grasping the meaning of the message. Develop in yourself the ability to speak distinctly; produce the sounds in a crisp and lucid manner.

Pronunciation

If articulation means speaking out all the sounds distinctly, then pronunciation requires us to speak out sounds in way that is generally accepted. The best way is to follow British Received pronunciation. One should be careful enough to pronounce individual sounds along with word stress according to the set norms. Wherever there is confusion, always consult a good dictionary and try to pronounce it correctly.

7. Voice Modulation

Modulation refers to the way we regulate, vary or adjust the tone, pitch, and volume of the sound or speaking voice. Modulation of voice brings flexibility and vitality to your voice, and you can express emotions and sentiments in the best possible way.

8. Pauses

A pause is a short silence flanked by words. A pause is speaking lets the listener reflect on the message and digest it accordingly. It helps you glide from one thought to another one. It embellishes your speech because it is a natural process to give a break.

Kinesics

Kinesics is the name given to the study of the body's physical movements. Nodding your head, blinking your eyes, shrugging shoulders, waving the hands, and other such physical activities are all forms of communication. Some kinesic behaviors are deliberately intended to communicate, as when your head for acceptance. Understanding non-verbal cues will develop your ability to use them more effectively during your presentations.

In face to face communication the message is conveyed on two levels simultaneously. One is verbal and other is non verbal. It is said by Socrates "Nobility and dignity; self-abasement and servility, insolence and vulgarity, are reflected in the face and in the attitudes of the body whether still or in motion". Non verbal communication is instinctive.

Non verbal communication is concerned with body movements (Kinesics), space (Proxemics) and vocal features (paralinguistic features). It is important that we know more about these features of body language because the verbal components of oral communication carry less than 35% of the social meaning of the situation, while more than65% is carried on the non-verbal band. People react strongly on what they see.

Kinesics includes:

Personal Appearance

Personal appearance plays an important role; people see you before they hear you. As you adapt your language to an audience, you should also dress appositely. Appearance includes clothes, hair, jewellery, cosmetics, and so on. In today's society the purpose of clothing has altered from fulfilling a basic need to expressing oneself. Clothes accent the body's movements.

Posture

Posture generally refers to the way we hold ourselves when we stand, sit or walk. The way you sit, stand, or walk reveals a lot about you. New speakers are unsure of what to do with their body. Certain mannerisms creep in, usually from nervousness - pacing constantly, bobbing the shoulders, fidgeting with notes, jingling coins, either constantly moving or remaining static. Truly, what one speaks is very important but you do just before you begin and after you have finished is equally important.

Gesture

Gesture is the movement made by hands, arms, shoulders, head and torso. Adroit gestures can add to the impact of speech. Gestures clarify your ideas or reinforce them and should be well suited to the audience and occasion. It has been observed that there are as many as 7,00,000 varied hand gestures alone and the meanings derived from them may vary from individual to individual.

Gestures should not divert the attention of the listener and distract from your message. Speakers' gestures should be quite natural and spontaneous.

Facial Expression

Facial expressions also play an important role in presentation. The face is the most expressive part of our body. A smile stands for friendliness, a frown for discontent, raised eyebrows for disbelief, tightened jaw muscles for antagonism, etc. The face rarely sends a single message at a time. Instead it sends a series of messages-your facial expressions may show anxiety, recognition, hesitation, and pleasure in quick succession.

Eye Contact

The eyes are considered to be the windows of the soul. You look to the eyes of a speaker to help find out the truthfulness of his speech, his intelligence, attitudes, and feelings. Eye contact is a direct and powerful form of non-verbal communication. Looking directly at listeners builds rapport. Prolonging the eye contact for three to five seconds tells the audience that you are sincere in what you say and that you want them to pay attention.

Proxemics

Proxemics is the study of physical space in interpersonal relations. The way people use space tells you a lot about them. In a professional setting, space is used to signal power and status. Your gestures should be in accordance with the space available.

Edward T Hall is his "The Hidden Dimension" divides space into four distinct zones.

1. Intimate

This zone starts with personal touch and extends just to 18 inches. Members of the family, relatives etc fall under this zone. The best relationship that describes it is the mother-child relationship.

2. Personal

This zone stretches from 18 inches to 4 feet. Your close friends, colleagues, peers etc fall in this group. It permits spontaneous and unprogrammed communications.

3. Social

Social events take place in this radius of 4 feet to 12 feet. In this layer, relationships are more official. People are more cautious in their movements. You should be smart enough to conduct it with less emotion and more planning.

4. Public

This zone starts from 12 feet and may extend to 30 feet or to the range of eyesight and hearing. Events that take place in this zone are formal. The audience is free to do whatever it feels like.

Chronemics

Chronemics is the study of how human beings communicate through their use of time. We attempt to control time, trying to use it more effectively. Good timing is very crucial, and you should rehearse a formal presentation until it is a little under line, because staying within time limits is a mark of courtesy and professionalism.

Inference/Conclusion

To recapitulate, oral presentation is an art that requires careful planning, preparation, and a great deal of practice. The tool is both valuable and relevant. Apart from communicating the main purpose of your presentation, there are various factors that you need to blend inextricably to convey your message clearly. These features are: audience analysis, organization of matter, preparation of an outline, nuances of delivery, kinesics and paralinguistic, and visual aids. Your aim should be to keep all these threads intact, neither too loose nor too tight. With care and practice, you can achieve wonders with your oral presentations.

Haptics (Touch):

Definition: Haptics refers to the use of touch to communicate emotions or convey meaning, such as handshakes, pats on the back, or high-fives. While touch can build rapport, it should be used appropriately in professional settings. A handshake, for instance, can convey respect and confidence in an engineering environment. Offering a handshake at the beginning or end of a meeting with a client or a colleague.

Olfactics (Smell):

Definition: Olfactics refers to the use of smell to communicate, though this is less often discussed in communication studies. In professional settings, smells can influence perceptions and interactions.

Importance in Engineering: Engineers should be aware of their personal hygiene and how strong scents (e.g., perfume, food) can affect the workplace atmosphere, especially in confined environments. Avoiding overpowering perfumes or food odors when attending meetings or working in shared spaces.

Oculesics (Eye Contact):

Definition: Oculesics refers to the use of eye contact to communicate emotions, attention, and engagement.

Importance in Engineering: Engineers need to maintain appropriate eye contact during presentations or meetings to convey confidence and sincerity. During a team meeting, maintaining eye contact with colleagues shows attentiveness and respect.